

ANDREW JACOBS BILSON

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PROFESSIONAL SUMMARY

Driven clinical pharmacist and emerging neuroscience researcher with multidisciplinary experience in neuropharmacology, pharmacogenomics, and clinical research. Skilled in integrating molecular and computational approaches to investigate neurodegenerative and psychiatric disorders. Passionate about advancing precision medicine and translational neuroscience to improve patient outcomes in brain health.

EDUCATION

Doctor of Pharmacy (PharmD) GPA: 3.78/4.00 (Class rank: 16/305, Top 5%) **Sept 2018 – Aug 2024**

Kwame Nkrumah University of Science and Technology

Kumasi, Ghana

Faculty of Pharmacy and Pharmaceutical Sciences

- Relevant Coursework: Systems Pharmacology I, II & III, Toxicology, Biochemistry I & II, Medicinal Chemistry I, II & III, Pharmaceutical Chemistry I & II, Analytical Chemistry I, II & III, Phytochemistry, Pharmacognosy, Molecular Biology I & II, Microbiology I & II, Biostatistics and Research Methods I & II, Public Health, and Pharmacoepidemiology.

RESEARCH / WORK EXPERIENCE

Pandam Laboratory, Kumasi Center for Collaborative Research in Tropical Medicine (KCCR)

Trainee Researcher (20 hours/week)

Jan 2026 – Present

- Engaged in bioinformatics and computational biology projects focusing on the integration of genomic and transcriptomic datasets to identify molecular signatures associated with neurological disorders, under the counsel of Dr. Salifu Pandam.
- Applying machine learning and statistical modeling to uncover genotype-phenotype correlations, contributing to the broader goal of advancing precision neuroscience research in African populations.

Pharmacogenomics Global Research Network (PGRN)

Research Contributor

June 2025 – Present

- Contributing to Parkinson's disease pharmacogenomics research as a trainee under the supervision of Professor Rachel Tyndale through exploration of genomic variant identification across populations, leveraging global datasets to inform precision medicine approaches.

Global Parkinson's Genetics Program (GP2)

Trainee Researcher

May 2025 – Present

- Advancing genomic analysis of Parkinson's disease cohorts, contributing to variant discovery and population-level comparisons to support inclusive genetic insights for under-represented regions, under the guidance of Drs. Vida Obese and Yasser Mecheri.

Komfo Anokye Teaching Hospital, Department of Research & Development

Kumasi, Ghana

Clinical Research Assistant (20 hours/week)

Jan 2025 – Present

- Involved in the Transforming Parkinson's Care in Africa (TraPCAf) study and assist in collection of demographic and clinical data, and biological samples (urine, blood, and stool) from study participants to contribute to improving data quality.
- Supported the setup and ongoing data collection for a hospital-based Movement and Neurodegenerative Disorder registry for tracking clinical patterns and outcomes in neurodegenerative diseases under the guidance of Dr. Vida Obese and Professor Fred Stephen Sarfo.
- Assisting in patient data collection and entry for a colorectal surgery registry and contributing to improving surgical care quality under the guidance of Dr. Kwabena Agbedinu and Professor Gifty Kwakye.

Undergraduate Research Thesis

Mar 2022 – Sept 2023

- Thesis topic: Investigating The Anti-Allergic Rhinitis Effect of Stigmasterol Against Toluene-2,4-diisocyanate (TDI)-Induced Rhinitis in Rats.
- Findings suggested anti-inflammatory and immunomodulatory properties, supporting stigmasterol's potential as a candidate for allergic airway disease treatment.
- Advisor: Dr. Aaron Opoku-Antwi.

MANUSCRIPTS UNDER REVIEW & IN PREPARATION

- Obese V., Domeh P., Akassi J., Agnenorku M., Adamu S., Ampofo B. K., Opare-Addo P. A., Asare-Ankomah K., Ababio H., **Bilson A. J.** & Sarfo F. S. (2025). *First Case of Creutzfeldt - Jakob disease in Ghana: Diagnostic Challenge in a Resource-Limited Setting. (Manuscript Under Review: Case Reports In Neurology Journal, Submitted December 2025).*
- Obese V., Nkum B., Akassi J., Agnenorku M., Adamu S., Opare-Addo P. A., **Bilson A. J.**, Ababio H., Rizig M. & Sarfo F. S. (2025). *Clinical and Demographic Predictors of Cognitive Performance among Parkinson's disease Patients in Ghana: Insights from a Hospital-Based Cohort. (Manuscript Under Review: Equity Neuroscience Journal, Submitted November 2025).*
- Obese V., Nkum B., Akassi J., Agnenorku M., Adamu S., Opare-Addo P. A., **Bilson A. J.**, Ababio H., Rizig M. & Sarfo F. S. (2025). *Clinical Characteristics and Predictors of Disease Progression in Parkinson's disease in a Cohort of Patients in the Ashanti Region, Ghana. (Manuscript Under Review: Equity Neuroscience Journal, Submitted October 2025).*
- Aderinto N., Ukoaka B.M., Yusuf I., Oluwakorede A.T., Hasan A., Babalola A.M., Bukky S.O., **Bilson A.J.**, Moardeyo A., Olatunji G., Kokori E. & Abraham I.C. (2025). *Screen Time Usage and Neurodevelopmental Conditions: A Scoping Review of Emerging Evidence. (Manuscript Under Review: International Journal of Mental Health and Addiction, Submitted July 2025).*
- Aderinto N., Ukoaka B.M., Yusuf I., Oluwakorede A.T., Hasan A., Babalola A.M., Bukky S.O., **Bilson A.J.**, Moardeyo A., Olatunji G., Kokori E. & Abraham I.C. (2025). *Comparison of Bridging Therapy versus Direct Endovascular Thrombectomy in Patients with Acute Ischemic Stroke Due to Large Vessel Occlusion: A Systematic Review of Randomized Controlled Studies. (Manuscript Under Review: Frontiers in Neurology Journal, Submitted June 2025).*
- Botchway B. & **Bilson A.J.** (2025). *Developing Personalized Rehabilitation Protocols Using Wearable Sensors and Machine Learning in Patients with Stroke. (In Preparation).*
- Badu-Bonsu P., Oppong K. E., Owusu-Antwi R., Omuojine J. P., **Bilson A.J.** & Owusu Aboagye G. (2025). *Assessing Caregivers Burden in the Management of Schizophrenia and Bipolar Disorder at a Teaching Hospital in Ghana. (In Preparation).*

SKILLS AND COMPETENCIES

- *Programming:* Python (NumPy, Matplotlib), R, MATLAB, OCTAVE
- *Machine Learning/ Data Science:* Scikit-learn
- *Operating Systems:* Linux, Windows, Mac
- *Data management:* REDCap, KoboToolbox
- *Document Creation:* Microsoft Office Suite, Mendeley, Zotero
- *Modeling Software:* ChemDraw, BioRender
- *Laboratory:* Murine model experimentation, Drug extraction and formulation, ELISA and biomarker analysis.
- *Languages:* English (Native), French (Novice Proficiency)

LICENSE AND PROFESSIONAL DEVELOPMENT CERTIFICATIONS

- Registered Pharmacist License, Pharmacy Council, Ghana. Issued July 2025. Credential ID: HPA 7119.
- Aspire Leaders Program (Aspire Institute)
- Data Management for Clinical Research (Vanderbilt University – Coursera)
- Data Analysis (S. P. Jain Institute of Management and Research – Coursera)
- Medical Neuroscience (Duke University – Coursera)

- Computational Neuroscience (University of Washington – Coursera)
- Fundamental Neuroscience for Neuroimaging (John Hopkins University – Coursera)

SELECTED LEADERSHIP AND VOLUNTEERING SERVICES

- **Content Manager, *Brainwaves with Dr. Vida*** Nov 2025 – Present
 - Develop and manage Brainwaves content from concept to publication, translating complex neuroscience and mental health topics into clear, engaging, and accessible narratives.
 - Oversee content planning, editorial direction, and publication schedules in collaboration with Dr. Vida Obese, ensuring consistency in tone, accuracy, and alignment with Brainwaves' educational mission. (Link: <https://thebrainwaves.health/team/>)
- **Executive Assistant, *Anidaso Parkinson's Disease Foundation*** Jun 2025 – Present
 - Coordinate and support community awareness and patient-centered outreach initiatives, with a focus on education, advocacy, and quality-of-life improvement for individuals living with Parkinson's disease.
 - Led and contributed to advocacy and public education efforts that addressed stigma around Parkinson's disease and promoted earlier health-seeking behavior among patients and caregivers. (Link: <https://www.anidasopd.org/>)
- **Volunteer, *Lemon Foundation*** Mar 2025 – Present
 - Participate in mental health advocacy and youth empowerment programs that promote emotional resilience, self-awareness, and access to psychosocial support. (Link: <https://lemonfoundationgh.org/>)

SELECTED CONFERENCES, WORKSHOPS AND SEMINARS

- Global Parkinson's Genetics Program (GP2) Bioinformatics Workshop, Accra-Ghana. August 2025.
- Alzheimer's Association's AAIC Neuroscience Next Virtual Conference. Conference on cutting-edge research and emerging trends in neurodegeneration, biomarkers, and therapeutic innovations. February 2025.
- Labroots Global Seminar Series (Precision Medicine, Cannabis Sciences, Organoid Research, Biomarker Discovery, Multiomics in Brain Disease). January 2025.
- Systematic Review and Meta-Analyses Workshop – African Community for Systematic Reviews & Meta-Analyses (ACSRM). July – August 2024.

SELECTED HONOURS / AWARDS

- Ghana National Scholarship Secretariat Scholar, 2022, 2023 & 2024 (Awarded based on academic merit and financial need).

PROFESSIONAL MEMBERSHIPS

- Michael J. Fox Foundation (MJFF) Data Community of Practice Sept 2025 – Present
- American Society for Clinical Pharmacology and Therapeutics (ASCPT) June 2025 – Present
- Alzheimer's Association Feb 2025 – Present
- International Parkinson's and Movement Disorder Society Feb 2025 – Present
- International Youth Neuroscience Association Jan 2025 – Present
- Pharmaceutical Society of Ghana (PSGH) Nov 2024 – Present
- Students' Academic Board, Faculty of Pharmacy and Pharmaceutical Sciences Jan 2021 – Aug 2024

REFERENCES

- **Prof. Kwesi Boadu Mensah**; Associate Professor, Department of Pharmacology, College of Health Sciences, Kwame Nkrumah University of Science and Technology, Ghana; Email: kbmensah.pharm@knust.edu.gh; Phone: (+233) 20 844 3173; Relationship: Mentor
- **Prof. Gifty Kwakye**; Associate Professor, Division of Colorectal Surgery, Michigan Medicine, University of Michigan, USA; Email: gkwakye@med.umich.edu; Phone: +1 (203) 887-0670; Relationship: Mentor